

Adarsh PATIL

CONTACT INFORMATION

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PROFILE SUMMARY

I have pursued research at both target-oriented industry labs and blue-skies academic research. My research contributions and publications aim to holistically design next generation systems, specifically memory systems for various application domains and compute paradigms - HPC, AI, serverless. My ethos of application driven hardware research is reflected in all my works, each of which span several aspects of my proficiency

I possess effective technical communication skills and have consistently proven my ability to meet deadlines and achieve project objectives. I work well in a team as well as independently. I am a constant learner, striving to expand my sphere of understanding. My education and industry experience has helped me develop technical capabilities in several adjacent areas within computer systems - operating systems, compilers, JIT libraries, network interconnect, analytical and mathematical modeling.

EDUCATION

- AUGUST 2023 DOCTOR OF PHILOSOPHY
University of Edinburgh, United Kingdom [ARM Fellowship]
Thesis: *Co-designing reliability and performance for datacenter memory*
Advisor: Prof. Vijay Nagarajan
- JULY 2017 M.TECH. (RESEARCH)
Indian Institute Of Science (IISc), Bangalore, India
Thesis: *Heterogeneity Aware Shared DRAM Cache for Integrated Heterogeneous Architectures*
GPA: 6.33/8.0 - *magna cum laude*
Advisor: Prof. R Govindarajan
- MAY 2012 BACHELOR OF ENGINEERING
Ramaiah Institute of Technology (MSRIT), Bangalore, India
GPA: 9.40/10.0 - *summa cum laude*

PROFESSIONAL EXPERIENCE

Dec 2023 - Current	Staff Research Engineer (Oct '25) Senior Research Engineer Systems architecture & technology group	ARM, Cambridge, UK
Apr 2023 - Nov 2023 (8 months)	Postdoctoral Research Associate Algorithms for next generation cloud systems	UNIVERSITY OF EDINBURGH, UK
Aug 2017 - Apr 2019 (1 year 8 months)	Research Scientist HPC Ecosystem and Applications Team	INTEL, Bangalore, India
Jun 2012 - Jul 2014 Jun 2011 - Aug 2011 (2 years 4 months)	Technology Analyst Summer Analyst Core Platforms, Data Center Engineering	GOLDMAN SACHS, Bangalore, India

SELECTED RESEARCH PUBLICATIONS

C3PO @ ICS '25	Improving compiler support for SIMD offload using Arm Streaming SVE
HCDS @ ASPLOS '24	UDON: Offloading to general purpose compute on CXL memory
DSN '23	Āpta: Fault-tolerant object-granular CXL disaggregated memory for accelerating FaaS https://adar.sh/apta
ISCA '21	Dvé: Improving DRAM Reliability and Performance On-Demand via Coherent Replication https://adar.sh/dve
TACO '17 Best Poster EECS 2017, Presented HiPEAC 2018	HAShCache: Heterogeneity-Aware Shared DRAMCache for Integrated Heterogeneous CPU-GPU Systems https://adar.sh/hashcache-taco
Arm / UEd Conference 2021	Improving Reliability and Performance of Datacenter Systems via Coherence https://adar.sh/arm-ed-conf-2021
UK Systems 2021	FaaS with CXL Disaggregated Shared Memory

PROJECTS

PROJECTS AT UNIV OF EDINBURGH	Redesigning datacenter co-ordination services for next generation hardware • Novel hardware primitives for disaggregated memory • Employing new algorithms for synchronization and scheduling Achieving persistence through replicated disaggregated memory
PROJECTS AT INTEL	Research into high-performance software, hardware, and systems for AI • Optimized compute libraries for neural networks - libxsmm, MKL-DNN • Architecture specific parallel algorithms - register blocking, vectorization • Datatypes for matrix representation - low-precision, sparsity • Hardware architecture design proposals with a deep understanding of underlying algorithms
PROJECTS AT INDIAN INSTITUTE OF SCIENCE	Address translation overheads in next generation x86 processors • TLB and Pagewalk performance in multicore architectures with large Die-Stacked DRAM Cache [Tech Report 2015, arXiv] https://adar.sh/caffe-compiler-optimize Program flow prediction in mobile devices • Accuracy of hardware branch predictor in ARM processors running Android https://adar.sh/BranchPredAndroid Compiler optimization transforms to improve performance on CPUs • Harris Corner Detection: https://adar.sh/compiler-optimize • Caffe Neural Networks: https://adar.sh/caffe-compiler-optimize Dynamic Scoping for C Language in Clang compiler https://adar.sh/VarMutate Database query optimization: Selectivity estimation of predicates in queries • ESS Dimensions Reduction for Plan Bouquet https://adar.sh/PlanIkebana

PROJECTS AT GOLDMAN SACHS Architect, design and implement solutions of various virtualization & linux technologies spanning datacenter compute, storage, networking

Hardware and OS Performance Benchmarking & Analysis

- Authored an automated benchmarking framework to run and report performance by running test suites on VMs and Baremetals
- Performance analysis & tuning for specialized internal apps (e.g. low Latency, high I/O, memory, network intensive)
- Validation using platform testing suites – SpecJBB, kmake, blacksholes, Dhrystone, Whetstone, Hackbench, Disk tests, Network uperf, lat proc

Linux Containers

- Architecting and implementing Containers for Goldman Sachs Cloud
- Possess a good understanding of underlying technology Namespaces, Cgroups, SELinux, Network configuration, Libvirt API

Thin client desktop VDI solution

- Engineered a Minimized and locked down Linux based solution
- Authored several PyGTK and X11 based applications for remote management, diagnostics, troubleshooting and NEA
- Network booted, kickstart and preseed based unsupervised install
- Engineered a stateless RAM-based network booted system on ARM based hardware

Engineering Nested Virtualization (Bromium vSentry) as a security solution

Vendor Interaction and liaising – Intel, VMware, Redhat

PROJECTS AT
M S RAMAIAH
INST OF TECH

Spoken language identification using machine learning
[Bachelor's dissertation] <https://adar.sh/spokenlang>

SNIDS: An Intelligent & Multiclass Support Vector Machines Based NIDS
[ICECIT 2012] <https://adar.sh/S-NIDS>
funded by Defense Research and Development Organization (DRDO), India

Line Birds (game) using OpenGL <https://adar.sh/linebird>

A parallel algorithm for Max Flow Algorithm using Ford-Fulkerson method
Lead developer of a student focused Linux Distro “ANDROMEDA Linux”

PROFESSIONAL ACTIVITIES

- IEEE Senior Member, December 2025
- ACM SIGARCH content editor, March 2024 - Present
- Web chair HPCA 2024
- Informatics Science Communication Group, Dec 2021 - Nov 2023
- ICSA@Informatics social media communication, Sept 2021 - Nov 2023
- Teaching assistant/Tutor INF2C-CS, University of Edinburgh, Aug 2019 - Dec 2019
- Student System Admin at CSA Department, IISc, Aug 2014 - Dec 2016
- Teaching Associate for the CUDA Teaching Centre, RIT, Jan 2012 - May 2012
- Chairman of VRGLINUX (GNU/Linux users group at RIT),
Secretary and member of executive committee of IEEE, RIT
Influential Member of RoboMSR, CodeMSRIT, Assoc. of Computer Engineers, 2011-12

ACHIEVEMENTS AND AWARDS

- Founding trustee of Dr. M R Gorbhal Foundation - a charitable organization which aims to promote research in Physics (2022)
- Best Poster at Electrical Science Divisional Symposium at IISc (2017)
- Certificate of distinction - Data Science Course, Johns Hopkins University on Coursera (2014)
- *Best outgoing achiever* - Dept. of CSE at MSRIT, (2012)
- First Place at National Level Project Competition held at RIT (2012)
- Second Place at "Random Hacks of Kindness #2" hackathon (2010)

MISCELLANEOUS

STRENGTHS

- Adaptability, Quick learner, Hardworking and Dedication
- Effective communicator and good leadership skills
- Always updated with latest technology and trends of market.
- Analytical and mathematical problem solving ability

HOBBIES

- Avid endurance athlete: 2 full and 18 half marathons, 1 HYROX event
- Hiking enthusiast - 25 Munros, several Corbetts, coastal and trail walks
- Blogging and research communication
- Organizer: HPCA 2024, TEDx, Pycon India, Random Hacks of Kindness

OTHER LINKS github.com/adarshpatil
in.linkedin.com/in/adarshpatil

REFERENCES **Academic References**
Vijay Nagarajan, PhD Advisor
Professor, University of Edinburgh / University of Utah
vijay.nagarajan@ed.ac.uk, vijay@cs.utah.edu

Prof. R Govindarajan, Master's Advisor
Professor, IISc
govind@serc.iisc.ernet.in

Industry References
Nikos Nikoleris
Senior Principal Engineer, Arm

Bharat Kaul
Director, Intel Parallel Computing Lab

KEYWORDS

✓ Academic research (PhD) ✓ Industrial research experience ✓ Memory architecture ✓ System design ✓ Architectural modeling and simulation ✓ Reliability, availability and serviceability (RAS) ✓ Coherence protocols ✓ CXL, OpenCAPI ✓ Disaggregated memory ✓ Serverless computing ✓ CPU, GPU, SoC architecture ✓ DRAM, HBM, HMC, DDR memory